Part A: Conceptual Understanding (Theory)

1. What is Logistic Regression and why is it suitable for classification?

Logistic Regression is a supervised machine learning algorithm used for classification problems. It predicts the probability of an observation belonging to a particular class using the sigmoid function.

The sigmoid function converts output values into probabilities between 0 and 1.

Why suitable for classification?

- Predicts probabilities of class membership.
- Works well for binary classification problems.
- Easy to interpret and implement.
- Computationally efficient.
- Provides good baseline performance.

In this project, Logistic Regression is used to classify customers as:

- 0 → Low Risk
- 1 → High Risk

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2. Explain classification performance metrics and why accuracy alone is insufficient.

Classification performance metrics are used to evaluate how well a classification model predicts target classes.

Common metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

Why Accuracy Alone Is Insufficient

Accuracy measures the percentage of correct predictions.

Formula:

Accuracy=TP+TNTP+TN+FP+FNAccuracy = \frac{TP + TN}{TP + TN + FP + FN}Accuracy=TP+TN+FP+FNTP+TN

In imbalanced datasets, accuracy can be misleading.

Example:

If 95% customers are Low Risk and only 5% are High Risk, a model predicting all customers as Low Risk will achieve 95% accuracy but fail to identify High Risk customers.

Therefore, metrics like Precision, Recall, F1-Score, and ROC-AUC are necessary for proper evaluation.

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3. Define Type-I Error and Type-II Error in the context of risk prediction.

Type-I Error (False Positive)

A Low-Risk customer is incorrectly classified as High-Risk.

Actual: Low Risk (0)

Predicted: High Risk (1)

Consequences:

- Good customers may be denied loans or services.
- Customer dissatisfaction.

Type-II Error (False Negative)

A High-Risk customer is incorrectly classified as Low-Risk.

Actual: High Risk (1)

Predicted: Low Risk (0)

Consequences:

- Bank may approve risky customers.
- Higher chance of fraud or default.

In banking applications, minimizing Type-II Error is generally more important.

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4. Explain Precision, Recall, F1-Score, TPR, and FPR.

Precision

Precision measures how many predicted positive cases are actually positive.

Formula:

Precision=TPTP+FPPrecision = \frac{TP}{TP + FP}Precision=TP+FPTP

High precision means fewer false alarms.

Recall (Sensitivity)

Recall measures how many actual positive cases are correctly identified.

Formula:

Recall=TPTP+FNRecall = \frac{TP}{TP + FN}Recall=TP+FNTP

High recall means fewer high-risk customers are missed.

F1-Score

F1-Score is the harmonic mean of Precision and Recall.

Formula:

F1=2(Precision×Recall)Precision+RecallF1 = \frac{2(Precision \times Recall)}{Precision + Recall}F1=Precision+Recall2(Precision×Recall)

It balances both Precision and Recall.

True Positive Rate (TPR)

TPR is another name for Recall.

Formula:

TPR=TPTP+FNTPR = \frac{TP}{TP + FN}TPR=TP+FNTP

False Positive Rate (FPR)

FPR measures how many negative cases are incorrectly classified as positive.

Formula:

FPR=FPFP+TNFPR = \frac{FP}{FP + TN}FPR=FP+TNFP

Lower FPR indicates better performance.

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5. What is AUC-ROC and how does it help in evaluating classifiers?

ROC Curve

ROC (Receiver Operating Characteristic) Curve plots:

- True Positive Rate (TPR)
- False Positive Rate (FPR)

at different classification thresholds.

AUC

AUC stands for Area Under the Curve.

The AUC score ranges from 0 to 1.

AUC Score	Interpretation
0.5	Random Prediction
0.6 – 0.7	Fair
0.7 – 0.8	Good
0.8 – 0.9	Very Good
>0.9	Excellent

Importance

- Measures classifier performance across all thresholds.
- Works well on imbalanced datasets.
- Helps compare multiple models.

A higher AUC indicates a better classifier.

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6. Why does imbalanced data create problems in classification models?

Imbalanced data occurs when one class has significantly more observations than another.

Example:

- Low Risk = 95%
- High Risk = 5%

Problems Caused by Imbalanced Data

1. Model becomes biased toward majority class.
2. Minority class predictions become poor.
3. High accuracy but low recall.

4. Important high-risk customers may be missed.
5. Increased Type-II Errors.

Solutions Used in This Project

- Under Sampling
- Over Sampling
- SMOTE (Synthetic Minority Oversampling Technique)
- ADASYN (Adaptive Synthetic Sampling)

These techniques balance the dataset and improve minority class prediction performance.

Conclusion

In this project, Logistic Regression was used as a baseline model and evaluated using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and AUC-ROC. Since the dataset was imbalanced, balancing techniques such as Under Sampling, Over Sampling, SMOTE, and ADASYN were applied. Decision Tree and Random Forest classifiers were then implemented and optimized using RandomizedSearchCV and GridSearchCV. Finally, ROC-AUC analysis was used to identify the best-performing model for high-risk customer prediction.